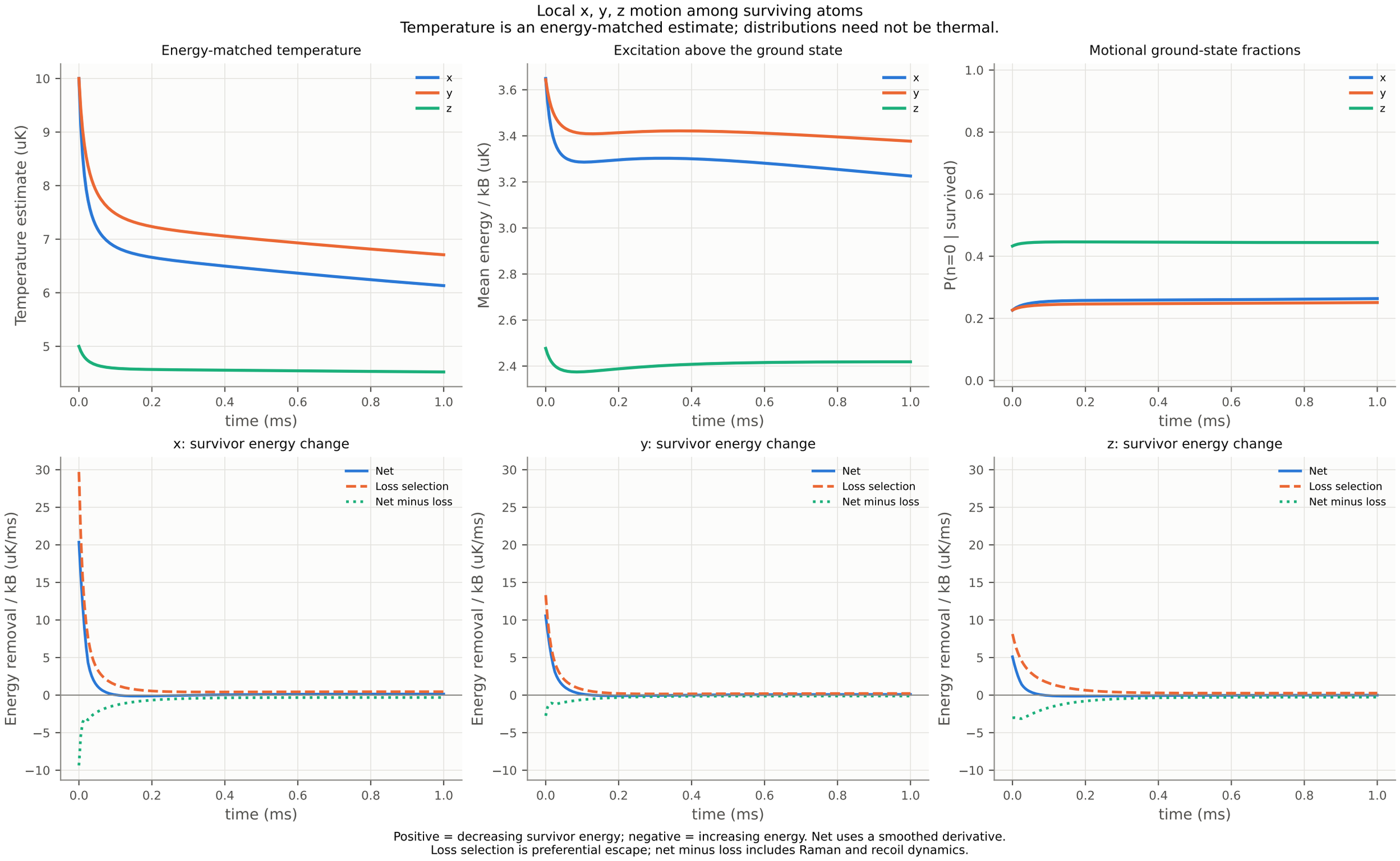
Local x, y, z motion among surviving atoms
Temperature is an energy-matched estimate; distributions need not be thermal.
Energy-matched temperature
Temperature estimate (uK)
Excitation above the ground state
Mean energy / kB (uK)
Motional ground-state fractions
P(n=0 | survived)
x
y
z
time (ms)
x: survivor energy change
y: survivor energy change
z: survivor energy change
Energy removal / kB (uK/ms)
Net
Loss selection
Net minus loss
Positive = decreasing survivor energy; negative = increasing energy. Net uses a smoothed derivative.
Loss selection is preferential escape; net minus loss includes Raman and recoil dynamics.